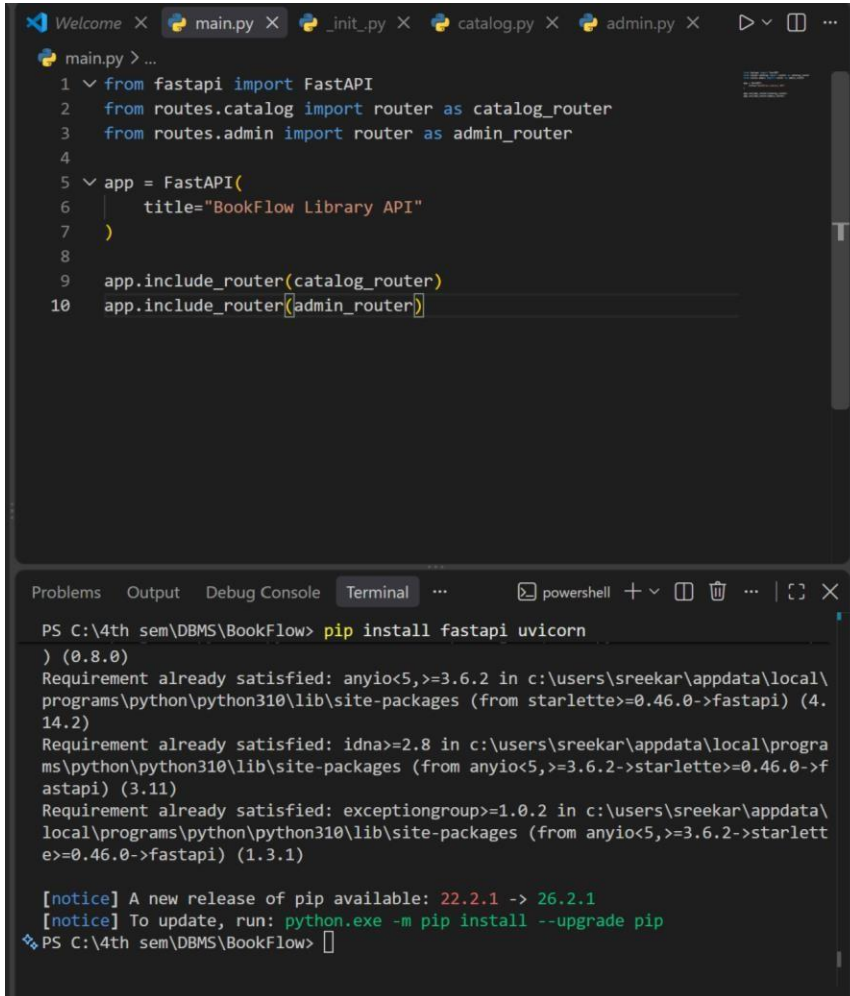


PRACTICAL 6



The image shows a VS Code editor with a Python project named 'BookFlow'. The editor has several tabs open: 'Welcome', 'main.py', '_init_.py', 'catalog.py', and 'admin.py'. The 'main.py' file is active and contains the following code:

```
1 from fastapi import FastAPI
2 from routes.catalog import router as catalog_router
3 from routes.admin import router as admin_router
4
5 app = FastAPI(
6     title="BookFlow Library API"
7 )
8
9 app.include_router(catalog_router)
10 app.include_router(admin_router)
```

Below the editor is a terminal window with the following output:

```
PS C:\4th sem\DBMS\BookFlow> pip install fastapi uvicorn
) (0.8.0)
Requirement already satisfied: anyio<5,>=3.6.2 in c:\users\sreekar\appdata\local\programs\python\python310\lib\site-packages (from starlette=>0.46.0->fastapi) (4.14.2)
Requirement already satisfied: idna>=2.8 in c:\users\sreekar\appdata\local\programs\python\python310\lib\site-packages (from anyio<5,>=3.6.2->starlette=>0.46.0->fastapi) (3.11)
Requirement already satisfied: exceptiongroup>=1.0.2 in c:\users\sreekar\appdata\local\programs\python\python310\lib\site-packages (from anyio<5,>=3.6.2->starlette=>0.46.0->fastapi) (1.3.1)

[notice] A new release of pip available: 22.2.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\4th sem\DBMS\BookFlow>
```


BookFlow Library API

0.1.0

OAS 3.1

/openapi.json

default

GET	/books	Get Books	⌵
GET	/books/{book_id}	Get Book	⌵
POST	/admin/books	Add Book	⌵
PUT	/admin/books/{book_id}	Update Book	⌵
DELETE	/admin/books/{book_id}	Delete Book	⌵

Schemas

Book > Expand all object

HTTPValidationError > Expand all object

ValidationError > Expand all object

```
> main.py X > init_.py X catalog.py X admin.py X
```

```
routes > admin.py > delete_book  
1 from fastapi import APIRouter, HTTPException  
2 from pydantic import BaseModel, Field  
3 from routes.catalog import books  
4  
5 router = APIRouter()  
6  
7  
8 class Book(BaseModel):  
9     title: str  
10    author: str  
11    price: float = Field(ge=0)  
12    isbn: str  
13    status: str  
14  
15  
16 @router.post("/admin/books")  
17 def add_book(book: Book):  
18     new_id = max([b["id"] for b in books], default=100) + 1  
19  
20     new_book = {
```

```
Problems Output Terminal ... powershell + - [ ] [X] X
```

```
PS C:\4th sem\DBMS\BookFlow> pip install fastapi uvicorn  
astapi) (0.8.0)  
Requirement already satisfied: anyio<5,>=3.6.2 in c:\users\sreekar\AppData\local\programs\python\python310\lib\site-packages (from starlette=>0.46.0->fastap  
i) (4.14.2)  
Requirement already satisfied: idna>=2.8 in c:\users\sreekar\AppData\local\pro  
grams\python\python310\lib\site-packages (from anyio<5,>=3.6.2->starlette=>0.4  
6.0->fastapi) (3.11)  
Requirement already satisfied: exceptiongroup>=1.0.2 in c:\users\sreekar\Appda  
ta\local\programs\python\python310\lib\site-packages (from anyio<5,>=3.6.2->st  
arlette=>0.46.0->fastapi) (1.3.1)
```

```
[notice] A new release of pip available: 22.2.1 -> 26.2.1  
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
❖ PS C:\4th sem\DBMS\BookFlow>
```

Request body required

application/json

Edit Value | Schema

```
{
  "title": "Machine Learning Basics",
  "author": "Alice Johnson",
  "price": 600,
  "isbn": "9781111111111",
  "status": "Available"
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/admin/books' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Machine Learning Basics",
    "author": "Alice Johnson",
    "price": 600,
    "isbn": "9781111111111",
    "status": "Available"
  }'
```

book_id * required
integer
(path)

Execute

Clear

Responses

Curl

```
curl -X 'GET' \  
'http://127.0.0.1:8000/books/101' \  
-H 'accept: */*' 
```

Request URL

```
http://127.0.0.1:8000/books/101
```

Server response

Code Details

200

Response body

```
{  
  "id": 101,  
  "title": "Python Programming",  
  "author": "John Smith",  
  "price": 450,  
  "isbn": "9781234567890",  
  "status": "Available"  
}
```



Download

Response headers

```
content-length: 117  
content-type: application/json  
date: Sun, 20 Sep 2026 18:32:16 GMT  
server: uvicorn
```

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/admin/books' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Machine Learning Basics",
    "author": "Alice Johnson",
    "price": 600,
    "isbn": "9781111111111",
    "status": "Available"
  }'
```

Request URL

http://127.0.0.1:8000/admin/books

Server response

Code Details

200

Response body

```
{
  "message": "Book added successfully",
  "book": {
    "id": 103,
    "title": "Machine Learning Basics",
    "author": "Alice Johnson",
    "price": 600,
    "isbn": "9781111111111",
    "status": "Available"
  }
}
```



Download

Response headers

```
content-length: 172
content-type: application/json
date: Sun, 20 Sep 2026 18:33:14 GMT
server: uvicorn
```

Responses

Name Description

book_id * required

integer
(path)

103

Request body required

application/json

Edit Value | Schema

```
{
  "title": "Advanced Machine Learning",
  "author": "Alice Johnson",
  "price": 750,
  "isbn": "9781111111111",
  "status": "Available"
}
```

Execute

Clear

Responses

Curl

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/books' \
-H 'accept: */*'

```

Request URL

http://127.0.0.1:8000/books

Server response

Code Details

200

Response body

```
[
  {
    "id": 101,
    "title": "Python Programming",
    "author": "John Smith",
    "price": 450,
    "isbn": "9781234567890",
    "status": "Available"
  },
  {
    "id": 102,
    "title": "Database Management",
    "author": "Robert Brown",
    "price": 550,
    "isbn": "9789876543210",
    "status": "Available"
  }
]
```



Download

Response headers

```
content-length: 240
content-type: application/json
date: Sun, 20 Sep 2026 18:30:45 GMT
server: uvicorn

```

Responses

Code Description

Links

Responses

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/admin/books/103' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Advanced Machine Learning",
    "author": "Alice Johnson",
    "price": 750,
    "isbn": "9781111111111",
    "status": "Available"
  }'
```

Request URL

http://127.0.0.1:8000/admin/books/103

Server response

Code

Details

200

Response body

```
{
  "message": "Book updated successfully",
  "book": {
    "id": 103,
    "title": "Advanced Machine Learning",
    "author": "Alice Johnson",
    "price": 750,
    "isbn": "9781111111111",
    "status": "Available"
  }
}
```



Download

Response headers

```
content-length: 176
content-type: application/json
date: Sun, 20 Sep 2026 18:34:42 GMT
server: uvicorn
```

Responses